

- Strukturna analiza digitalne datoteke

“

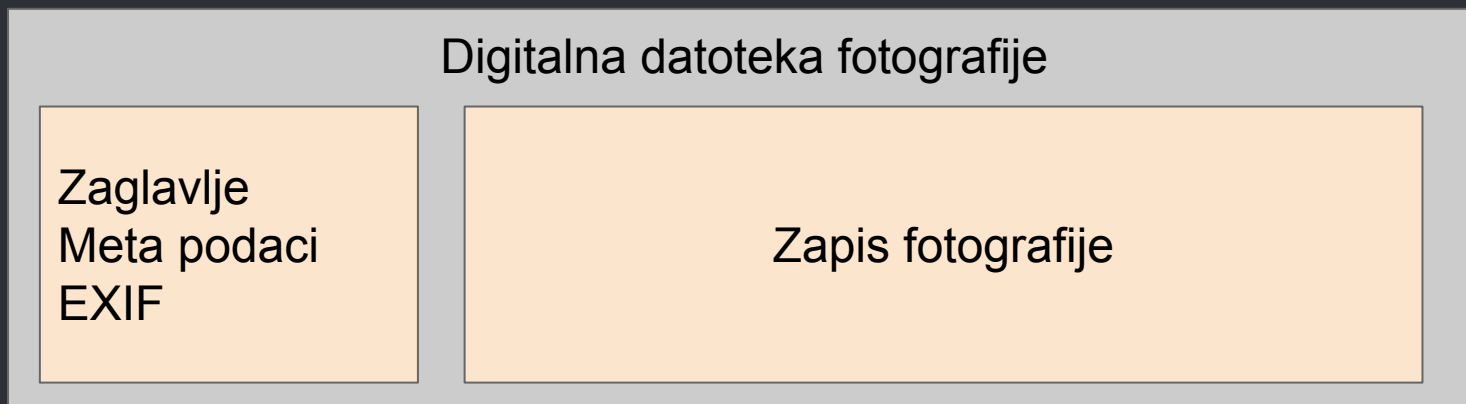
*Digitalna fotografija sadrži puno više
informacija nego što programi za
prikaz fotografije mogu prikazati*

1

Struktura datoteke digitalne fotografije

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● Struktura digitalne datoteke



- **Dodatne informacije (vezane uz datoteku):** ime datoteke, veličina datoteke, MAC vremena

Autentična digitalna datoteka : datoteka izvorna datoteka nastala u uređaju za snimanje (digitalnom fotoaparatu)

Autentična fotografija: izvorno snimljena fotografija bez ikakvih slučajnih ili namjernih promjena

Promjenom na digitalnoj datoteci, sama fotografija može i dalje ostati autentična

- MAC vremena

Modified, Access & Create time

- Vrijeme izmjene (mtime)

```
popaj /home/dkrst/nastava/NOVI_KOLEGIJI/Forenzika_slike/slike/work> stat DSC_0066.JPG
  File: 'DSC_0066.JPG'
  Size: 7271703      Blocks: 14208      IO Block: 4096   regular file
Device: 804h/2052d   Inode: 17569261    Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   dkrst)   Gid: ( 1000/   dkrst)
Access: 2017-10-11 17:31:41.950307056 +0200
Modify: 2017-10-02 13:40:21.141025041 +0200
Change: 2017-10-02 13:40:21.141025041 +0200
Birth: -
popaj /home/dkrst/nastava/NOVI_KOLEGIJI/Forenzika_slike/slike/work> stat DSC_0066-edited.JPG
  File: 'DSC_0066-edited.JPG'
  Size: 2371060      Blocks: 4632      IO Block: 4096   regular file
Device: 804h/2052d   Inode: 17569264    Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   dkrst)   Gid: ( 1000/   dkrst)
Access: 2017-10-11 17:32:44.791040458 +0200
Modify: 2017-10-02 13:45:24.148606655 +0200
Change: 2017-10-02 13:45:24.148606655 +0200
Birth: -
popaj /home/dkrst/nastava/NOVI_KOLEGIJI/Forenzika_slike/slike/work>
```

promjene meta podataka (ne mora se mijenjati
sadržaja datoteke) - **Change time**

- Windows: vrijeme nastanka digitalne datoteke

● Analiza heksadecimalnog zapisa

- Aplikacije za obradu digitalne slike ostavljaju trag u datoteci
- Ovi tragovi mogu uključivati razne informacije, od jednostavnog potpisa do popisa operacija izvedenih na fotografiji

Kontraforenzika:

- Prikrivanje tragova interakcije s digitalnom fotografijom editiranjem heksadecimalnog zapisa u editor

Pregledavanjem digitalne datoteke u heksadecimalnom editoru moguće je otkriti tragove krivotvorenja na fotografiji

00000001e	9
00000002d	0
00000003c	0
00000004b	0
00000005a	0
000000069	0
000000078	0
000000087	0
000000096	8
0000000a5	2
0000000b4	4
0000000c3	0
0000000d2	7
0000000e1	3
0000000f0	2
0000000ff	8
00000010e	0
00000011d	0
00000012c	0
00000013b	9

```
0000000a3 | 01 00 00 00 3c 01 0
```

000000d2	01	00	00	00	2C	01	00	00	01	00	00	00	47	49	4D	GIM
000000e1	50	20	32	2E	38	2E	31	36	00	32	30	31	37	3A	31	P 2.8.16.2017:1
000000f0	30	3A	30	32	20	31	33	3A	34	35	3A	31	36	00	28	0:02 13:45:16.(
000000ff	00	9A	82	05	00	01	00	00	00	C6	02	00	00	9D	82	
0000010e	05	00	01	00	00	00	CE	02	00	00	22	88	03	00	01	"
0000011d	00	00	00	00	00	00	00	27	88	03	00	01	00	00	00	'
0000012c	C8	00	00	00	00	90	07	00	04	00	00	00	30	32	33	023
0000013b	30	03	90	02	00	14	00	00	00	D6	02	00	00	04	90	0.....
0000014a	02	00	14	00	00	00	EA	02	00	00	01	91	07	00	04	
00000159	00	00	00	01	02	03	00	02	91	05	00	01	00	00	00	
00000168	FE	02	00	00	04	92	0A	00	01	00	00	00	06	03	00	
00000177	00	05	92	05	00	01	00	00	00	0E	03	00	00	07	92	
00000186	03	00	01	00	00	00	05	00	00	00	08	92	03	00	01	

```

.....JFIF.....
.....U9Exif..
.....
.....(
.....1.....
.....2.....
.....i
.....%
R;..d;..NIKO
CORPORATION.N
ON D5300..
2.8.16.2017:1

```

Search: 00005aa5 78 69 66 3a 52 65 7

Search	0000	0000	0000	0000
Un	0000	0000	0000	0000
S	0000	0000	0000	0000
Uns	0000	0000	0000	0000
☐ S	0000	0000	0000	0000
	0000	0000	0000	0000
	0000	0000	0000	0000
	0000	0000	0000	0000
	0000	0000	0000	0000

00005aa5	78	69	66	3A	52	65	73	6F	6C	75	74	69	6F	6E	55
00005ab4	6E	69	74	3E	0A	20	20	3C	65	78	69	66	3A	53	6F
00005ac3	66	74	77	61	72	65	3E	47	49	4D	50	20	32	2E	38
00005ad2	2E	31	36	3C	2F	65	78	69	66	3A	53	6F	66	74	77
00005ae1	61	72	65	3E	0A	20	20	3C	65	78	69	66	3A	44	61
00005af0	74	65	54	69	6D	65	3E	32	30	31	37	3A	31	30	3A
00005aff	30	32	20	31	33	3A	34	33	3A	33	38	3C	2F	65	78
00005b0e	69	66	3A	44	61	74	65	54	69	6D	65	3E	0A	20	20
00005b1d	3C	65	78	69	66	3A	59	43	62	43	72	50	6F	73	69
00005b2c	74	69	6F	6E	69	6E	67	3E	43	65	6E	74	65	72	65
00005b3b	64	3C	2E	65	78	69	66	3A	59	43	62	43	72	50	6F

```
xif:ResolutionUnit>. <exif:Software>GIMP 2.8.16</exif:Software>. <exif:DateTime>2017:10:02 13:43:38</exif:DateTime>. <exif:YCbCrPositioning>Centered</exif:YCbCrPo
```

Find Next Find Previous
 Hexadecimal: 20 32 2E 38
 Decimal: 032 050 046 056
 Octal: 040 062 056 070
 Binary: 00100000 00110010 00101
 ASCII Text: 2.8
 e to 0xe1 (0x4 bytes) INS

● Analiza heksadecimalnog zapisa

- Fotografija sadrži puno više informacija nego što programi za prikaz slike mogu prikazati
- Korištenjem heksadecimalnih editora informacija sadržana u datoteci digitalne fotografije preglednija je za analizu
 - Korištenjem standardnih funkcija, kao što je npr. pretraživanje omogućena je znatno jednostavnija analiza digitalnih datoteka

Heksadecimalni editori omogućavaju jednostavnu manipulaciju digitalnim fotografijama

- svaka greška može rezultirati oštećenom i nečitljivom datotekom!

● JPEG, JFIF, EXIF

JPEG (Joint Photographic Expert Group) standard specificira kodek

- Specificiran je način kodiranja ulazne slike (matrice piksela) u komprimirani stream bitova
- Nije specificiran format datoteke za pohranu streama bitova
 - Nije specificirana organizacija (redosljed) različitih dijelova JPEG streama (Y, Cb, Cr kanali se nezavisno komprimiraju)
 - Nije specificiran način pohrane drugih potrebnih podataka (razolucija, kvantizacijske tablice, Huffmanove tablice, ...)

JFIF (JPEG File Interchange Format) i EXIF (Exchangable Image File Format) standardi definiraju (najčešći) format datoteke za pohranu JPEG komprimiranih slika

- JIF (JPEG Interchange Format) - starija i rijetko korištena verzija standarda

JFIF

APP0

JFIF application marker segment

FF E0 nn nn 4A 46 49 46 00 ...

JFIF-APP0 marker segment

Polje	Veličina (bytes)	Opis
APP0 marker	2	FF E0 (oznaka početka segmenta)

JFXX-APP0 ekstenzija

Polje	Veličina (bytes)	Opis	ninirano)
APP0 marker	2	FF E0 (oznaka početka segmenta)	
nn nn	2	Dužina segmenta bez APP0 markera	
Identifikator	5	4A 46 58 58 00 (ASCII: JFXX, null-treminiran)	
Thumbnail format	1	10 - JPEG format 11 - 1 byte po pixel (paleta) 13 - 3 byte po pikelu RGB (nekomprimirano)	

Thumbnail data (ovisno o formatu)

EXIF

Exchangeable Image File Format (EXIF) je standard za pohranu informacija o parametrima nastanka fotografije

- EXIF specifikacija koristi postojeće formate datoteka na koje dodaje svoje meta podatke (JPEG, TIFF, audio podaci, ...)
- Meta podaci dodaju se na kodirani JPEG stream:
 - model fotoaparata
 - Vrijeme i datum nastanka fotografije
 - GPS lokacija (ako je dostupna)
 - otvor objektiva
 - trajanje ekspozicije
 - ISO osjetljivost
 - ...
 - umanjena (thumbnail) fotografija

Polja koja sadrži EXIF zapis nisu standardizirana

```

ExifTool Version Number      : 10.10
File Name                    : DSC_0066-b.jpg
Directory                   :
File Size                   :
File Modification Date/Time  :
File Access Date/Time       :
File Inode Change Date/Time  :
File Permissions             :
File Type                   :
File Type Extension         :
MIME Type                   :
Exif Byte Order             : Little-endian (Motorola)
Make                        : Canon
Camera Model                 : Canon EOS 5D Mark IV
Orientation                 : Horizontal (normal)
X Resolution                 : 72
Y Resolution                 : 72
Resolution Unit              : None
Software                     : Canon EOS
Modify Date                  : 2017:08:18 18:59:56
Y Cb Cr Positioning         : Default
Exposure Time                : 1/250
F Number                     : 3.5
Exposure Program             : P
ISO                           : 12800
Sensitivity                   : 12800
Exif Version                 : 2.3.0
Date/Time Original           : 2017:08:18 18:59:56
ISO2                          : 12800
Create Date                  : 2017:08:18 18:59:56
Components Configuration    : RGB, A, B, G
Compressed Bits               : 16
Exposure Compensation        : 0
Max Aperture                 : f/3.5
Metering Mode                : Multi-segment
Light Source                  : Flash
Flash                        : On
Focal Length                 : 34.0 mm
Maker Note Version          : 1.0
Quality                      : Standard
White Balance                : Auto
Focus Mode                   : Single
Flash Setting                : On
Flash Type                   : Built-in
White Balance Bracket        : Off
WB RB Levels                 : 0
Program Shift                : Off
Exposure Diff                : 0
Flash Exposure Lock          : Off
ISO Setting                  : 12800
Image Boundary               :
External Flash Exposure Comp : 0
Flash Exposure Bracketing    : Off
Exposure Bracketing          : Off
Crop Hi Speed                : Off
Exposure Time Bracketing     : Off
Serial Number                :
Image Size                   : 6000x4000
Shutter Count                : 1
Vibration Reduction          : Off
Flash Info Version           : 1.0
VR Mode                      : Off
Active D-Lighting            : Off
Picture Control              : Standard
Multi Exposure Version       : 1.0
Multi Exposure Mode          : Off
Multi Exposure Shots         : 1
Multi Exposure Auto Gain     : Off
High ISO Noise Reduction     : Off
Power Up Time                : 0
AF Info 2 Version           : 1.0
Contrast Detect AF           : Off
AF Area Mode                 : Single
Phase Detect AF              : Off
Primary AF Point             : 1
AF Points Used               : 11
Contrast Detect AF In Focus  : Off
File Info Version           : 1.0
Directory Number            : 1
File Number                  : 0066
Retouch Info Version         : 1.0
Retouch NEF Processing       : Off
User Comment                 :
Sub Sec Time                 : 0
Sub Sec Time Original        : 0
Sub Sec Time Digitized       : 0
Flashpix Version             : 1.0
Color Space                  : sRGB
Exif Image Width             : 6000
Exif Image Height            : 4000
Interoperability Index       : 1
Interoperability Version     : 1.0
Sensing Method               : 2
File Source                   : File
Scene Type                    : 0
CFA Pattern                  : 1
Custom Rendered              : Off
Exposure Mode                : P
Digital Zoom Ratio           : 1.0
Focal Length In 35mm Format  : 51.0 mm
Scene Capture Type           : Standard
Gain Control                  : None
Contrast                     : Normal
Saturation                   : Normal
Sharpness                    : Normal
Subject Distance Range       : Unknown
GPS Version ID               : 2.3.0.0
Compression                  : JPEG (old-style)
Thumbnail Offset             : 15872
Thumbnail Length             : 7839
Rating                       : 0
MPF Version                  : 0100
Number Of Images             : 3
MP Image Flags               : Dependent child image
MP Image Format               : JPEG
MP Image Type                : Large Thumbnail (full HD equivalent)
MP Image Length              : 476951
MP Image Start               : 6794752
Dependent Image 1 Entry Number : 0
Dependent Image 2 Entry Number : 0
Image Width                  : 6000
Image Height                  : 4000
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:2 (2 1)
Aperture                     : 8.0
Auto Focus                   : On
Blue Balance                  : 1.421875
Image Size                   : 6000x4000
Lens ID                      : Unknown (A0 40 2D 53 2C 3C CA 8E)
Lens                         : 18-55mm f/3.5-5.6 G VR [7]
Preview Image                 : (Binary data 35959 bytes, use -b option to display)
MP Image 3                   : (Binary data 476951 bytes, use -b option to display)
Megapixels                   : 24.0
Red Balance                   : 1.996094
Scale Factor To 35 mm Equivalent : 1.5
Shutter Speed                 : 1/250
Create Date                  : 2017:08:18 18:59:56.30
Date/Time Original           : 2017:08:18 18:59:56.30
Modify Date                  : 2017:08:18 18:59:56.30
Thumbnail Image              : (Binary data 7839 bytes, use -b option to display)
Circle Of Confusion          : 0.020 mm
Depth Of Field                : 5.29 m (2.48 - 7.77 m)
Field Of View                 : 38.6 deg (2.63 m)
Focal Length                  : 34.0 mm (35 mm equivalent: 51.0 mm)
Hyperfocal Distance          : 7.21 m
Light Value                   : 13.0
(End)

```

2

Forenzička analiza digitalne datoteke

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Koja slika je izvorna, a koja modificirana?



● Slika 1



Slika 1 - osnovne informacije o datoteci

```
popaj /home/dkrst/nastava/NOVI_KOLEGIJI/Forenzika_slike/slike/work> ls -al DSC_0066-1.jpg
-rw-rw-r-- 1 dkrst dkrst 7271703 Lis  2 14:00 DSC_0066-1.jpg
popaj /home/dkrst/nastava/NOVI_KOLEGIJI/Forenzika_slike/slike/work> stat DSC_0066-1.jpg
  File: 'DSC_0066-1.jpg'
  Size: 7271703          Blocks: 14208          IO Block: 4096    regular file
Device: 804h/2052d      Inode: 17569306       Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   dkrst)   Gid: ( 1000/   dkrst)
Access: 2017-10-12 15:24:59.788214705 +0200
Modify: 2017-10-02 14:00:52.662988967 +0200
Change: 2017-10-13 12:49:07.336404268 +0200
 Birth: -
popaj /home/dkrst/nastava/NOVI_KOLEGIJI/Forenzika_slike/slike/work> █
```

Ime datoteke:

DSC_0066-1.jpg

Veličina datoteke:

7271703 (6.935MB)

MAC vremena:

- Modification time: 14:00:52 02. 10. 2017.
- Access time: 15:24:59 12. 10. 2017.
- Creation (change) time: 12:49:07 13. 10. 2017.

- Slika 1 - analiza heksadecimalnog zapisa

SOI

APP1

Exif

```
DSC_0066-1.jpg x
00000000 FF D8 FF E1 FF FE 45 78 69 66 00 00 49 49 2A .....Exif..II*
0000000f 00 08 00 00 00 0B 00 0F 01 02 00 12 00 00 00 .....
0000001e 94 00 00 00 10 01 02 00 0C 00 00 00 A8 00 00 .....
```

model fotografskog
aparata

vrijeme

```
DSC_0066-1.jpg x
0000005a 02 00 00 00 31 01 .....1.....
00000069 00 32 01 02 00 14 .....2.....
00000078 03 00 01 00 00 00 01 00 00 00 89 87 04 00 01 .....i....
00000087 00 00 00 E4 00 00 00 25 88 04 00 01 00 00 00 .....%.....
00000096 84 3B 00 00 98 3B 00 00 00 00 4E 49 4B 4F 4E ;...;...NIKON
000000a5 20 43 4F 52 50 4F 52 41 54 49 4F 4E 00 00 00 CORPORATION...
000000b4 4E 49 4B 4F 4E 20 44 35 33 30 30 00 2C 01 00 NIKON D5300...
000000c3 00 01 00 00 00 2C 01 00 00 01 00 00 00 56 65 .....Ve
000000d2 72 2E 31 2E 30 31 20 00 00 00 32 30 31 37 3A r.1.01 ...2017:
000000e1 30 38 3A 31 38 20 31 38 3A 35 39 3A 35 36 00 08:18 18:59:56.
```

vrijeme (isto, ponavlja se 2 puta)

```
DSC_0066-1.jpg x
000002a3 A4 03 00 01 00 00 00 00 00 00 00 00 00 00 .....
000002b2 01 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
000002c1 00 00 00 00 00 0A A4 03 00 01 00 00 00 00 00 .....
000002d0 00 00 0C A4 03 00 01 00 00 00 00 00 00 00 .....
000002df 00 00 00 00 00 0A 00 00 00 00 C4 09 00 00 50 00 .....P.
000002ee 00 00 0A 00 00 00 32 30 31 37 3A 30 38 3A 31 .....2017:08:1
000002fd 38 20 31 38 3A 35 39 3A 35 36 00 32 30 31 37 8 18:59:56.2017
0000030c 3A 30 38 3A 31 38 20 31 38 3A 35 39 3A 35 36 :08:18 18:59:56
0000031b 00 02 00 00 00 01 00 00 00 00 00 00 00 06 00 .....
0000032a 00 00 2C 00 00 00 0A 00 00 00 54 01 00 00 0A ...T....
```

Slika 1 - analiza

```
ExifTool Version Number      : 10.10
File Name                    : DSC_0066-1.jpg
Directory                    : .
File Size                    : 6.9 MB
File Modification Date/Time  : 2017:10:02 14:
File Access Date/Time       : 2017:10:16 09:
File Inode Change Date/Time  : 2017:10:13 12:
File Permissions             : rw-rw-r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
Exif Byte Order              : Little-endian
Make                         : NIKON CORPORATION
Camera Model Name            : NIKON D5300
Orientation                  : Horizontal (normal)
X Resolution                  : 300
Y Resolution                  : 300
Resolution Unit              : inches
Software                     : Ver. 1.01
Modify Date                   : 2017:08:18 18:
Y Cb Cr Positioning          : Centered
Exposure Time                : 1/250
F Number                     : 8.0
Exposure Program             : Not Defined
ISO                          : 200
Sensitivity Type              : Recommended Exposure
Exif Version                 : 0230
Date/Time Original           : 2017:08:18 18:
Create Date                  : 2017:08:18 18:
Components Configuration    : Y, Cb, Cr, -
Compressed Bits Per Pixel    : 2
Exposure Compensation       : 0
Max Aperture Value           : 4.6
Metering Mode                : Multi-segment
Light Source                 : Unknown
Flash                       : Auto, Did not fire
Focal Length                 : 34.0 mm
Maker Note Version           : 2.11
Quality                      : Normal
White Balance                : Auto
Focus Mode                   : AF-A
```

```
Sensing Method               : One-chip color area
File Source                  : Digital Camera
Scene Type                   : Directly photographed
CFA Pattern                  : [Red,Green][Green,Blue]
Custom Rendered              : Normal
Exposure Mode                : Auto
Digital Zoom Ratio           : 1
Focal Length In 35mm Format  : 51 mm
Scene Capture Type           : Standard
Gain Control                 : None
Contrast                     : Normal
Saturation                   : Normal
Sharpness                    : Normal
Subject Distance Range       : Unknown
GPS Version ID               : 2.3.0.0
Compression                  : JPEG (old-style)
Thumbnail Offset             : 15872
Thumbnail Length             : 7839
Rating                       : 0
MPF Version                  : 0100
Number Of Images             : 3
MP Image Flags               : Dependent child image
MP Image Format               : JPEG
MP Image Type                : Large Thumbnail (full HD equivalent)
MP Image Length              : 476951
MP Image Start               : 6794752
Dependent Image 1 Entry Number : 0
Dependent Image 2 Entry Number : 0
Image Width                  : 6000
Image Height                 : 4000
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample              : 8
Color Components              : 3
Y Cb Cr Sub Sampling         : YCbCr4:2:2 (2 1)
Aperture                     : 8.0
Auto Focus                   : On
Blue Balance                 : 1.421875
Image Size                   : 6000x4000
Lens ID                      : Unknown (A0 40 2D 53 2C 3C CA 8E)
Lens                        : 18-55mm f/3.5-5.6 G VR [7]
Preview Image                : (Binary data 35959 bytes, use -b option to display)
MP Image 3                   : (Binary data 476951 bytes, use -b option to display)
Megapixels                   : 24.0
Red Balance                  : 1.996094
Scale Factor To 35 mm Equivalent: 1.5
Shutter Speed                : 1/250
Create Date                  : 2017:08:18 18:59:56.30
Date/Time Original           : 2017:08:18 18:59:56.30
Modify Date                  : 2017:08:18 18:59:56.30
Thumbnail Image              : (Binary data 7839 bytes, use -b option to display)
```

● Slika 2



Slika 2 - osnovne informacije o datoteci

```
popaj /home/dkrst/nastava/NOVI_KOLEGIJI/Forenzika_slike/slike/work> ls -al DSC_0066-2.jpg
-rw-rw-r-- 1 dkrst dkrst 2371060 Lis  2 14:00 DSC_0066-2.jpg
popaj /home/dkrst/nastava/NOVI_KOLEGIJI/Forenzika_slike/slike/work> stat DSC_0066-2.jpg
  File: 'DSC_0066-2.jpg'
  Size: 2371060          Blocks: 4632          IO Block: 4096   regular file
Device: 804h/2052d      Inode: 17569289      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/   dkrst)   Gid: ( 1000/   dkrst)
Access: 2017-10-16 09:36:22.539058671 +0200
Modify: 2017-10-02 14:00:42.962882540 +0200
Change: 2017-10-13 12:49:25.036643054 +0200
 Birth: -
popaj /home/dkrst/nastava/NOVI_KOLEGIJI/Forenzika_slike/slike/work> █
```

Ime datoteke:

DSC_0066-2.jpg

Veličina datoteke:

2371060 (2.26MB)

MAC vremena:

- Modification time: 14:00:42 02. 10. 2017.
- Access time: 09:36:22 16. 10. 2017.
- Creation (change) time: 12:49:25 13. 10. 2017.

DSC_0066-2.jpg

```
000056d6 64 66 3A 44 65 73 63 72 69 70 74 69 6F 6E 20 78 6D 6C 6E df:Description xmln
000056e9 73 3A 65 78 69 66 3D 27 68 74 74 70 3A 2F 2F 6E 73 2E 61 s:exif='http://ns.a
000056fc 64 6F 62 65 2E 63 6F 6D 2F 65 78 69 66 2F 31 2E 30 2F 27 dobe.com/exif/1.0/'
0000570f 3E 0A 20 20 3C 65 78 69 66 3A 4D 61 6B 65 3E 4E 49 4B 4F >. <exif:Make>NIKO
00005722 4E 20 43 4F 52 50 4F 52 41 54 49 4F 4E 3C 2F 65 78 69 66 N CORPORATION</exif
00005735 3A 4D 61 6B 65 3E 0A 20 20 3C 65 78 69 66 3A 4D 6F 64 65 :Make>. <exif:Mode
00005748 6C 3E 4E 49 4B 4F 4E 20 44 35 33 30 30 3C 2F 65 78 69 66 l>NIKON D5300</exif
0000575b 3A 4D 6F 64 65 6C 3E 0A 20 20 3C 65 78 69 66 3A 4F 72 69 :Model>. <exif:Ori
0000576e 65 6E 74 61 74 69 6F 6E 3E 54 6F 70 2D 6C 65 66 74 3C 2F entation>Top-left</
00005781 65 78 69 66 3A 4F 72 69 65 6E 74 61 74 69 6F 6E 3E 0A 20 exif:Orientation>.
00005794 20 3C 65 78 69 66 3A 58 52 65 73 6F 6C 75 74 69 6F 6E 3E <exif:XResolution>
000057a7 33 30 30 3C 2F 65 78 69 66 3A 58 52 65 73 6F 6C 75 74 69 300</exif:XResoluti
000057ba 6F 6E 3E 0A 20 20 3C 65 78 69 66 3A 59 52 65 73 6F 6C 75 on>. <exif:YResolu
000057cd 74 69 6F 6E 3E 33 30 30 3C 2F 65 78 69 66 3A 59 52 65 73 tion>300</exif:YRes
000057e0 6F 6C 75 74 69 6F 6E 3E 0A 20 20 3C 65 78 69 66 3A 52 65 olution>. <exif:Re
000057f3 73 6F 6C 75 74 69 6F 6E 55 6E 69 74 3E 49 6E 63 68 3C 2F solutionUnit>Inch</
00005806 65 78 69 66 3A 52 65 73 6F 6C 75 74 69 6F 6E 55 6E 69 74 exif:ResolutionUnit
00005819 3E 0A 20 20 3C 65 78 69 66 3A 53 6F 66 74 77 61 72 65 3E >. <exif:Software>
0000582c 56 65 72 2E 31 2E 30 31 20 3C 2F 65 78 69 66 3A 53 6F 66 Ver.1.01 </exif:Sof
0000583f 74 77 61 72 65 3E 0A 20 20 3C 65 78 69 66 3A 44 61 74 65 tware>. <exif:Date
00005852 54 69 6D 65 3E 32 30 31 37 3A 30 38 3A 31 38 20 31 38 3A Time>2017:08:18 18:
00005865 35 39 3A 35 36 3C 2F 65 78 69 66 3A 44 61 74 65 54 69 6D 59:56</exif:DateTim
00005878 65 3E 0A 20 20 3C 65 78 69 66 3A 59 43 62 43 72 50 6F 73 e>. <exif:YCbCrPos
```

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```
000002df 00 00 00 00 00 0A 00 00 00 C4 09 00 00 50 00
000002ee 00 00 0A 00 00 00 32 30 31 37 3A 30 38 3A 31
000002fd 38 20 31 38 3A 35 39 3A 35 36 00 32 30 31 37
0000030c 3A 30 38 3A 31 38 20 31 38 3A 35 39 3A 35 36
0000031b 00 02 00 00 00 01 00 00 00 00 00 00 06 00
```

```
.....P.
.....2017:08:1
8 18:59:56.2017
:08:18 18:59:56
```

● Slika 2 - a

ExifTool Version Number	: 10	Date Display Format	: D/M/Y	Sensing Method	: One-chip color area
File Name	: DS	ISO Expansion	: Off	File Source	: Digital Camera
Directory	:			Scene Type	: Directly photographed
File Size	: 2			CFA Pattern	: [Red,Green][Green,Blue]
File Modification Date/Time	: 20			Custom Rendered	: Normal
File Access Date/Time	: 20			Exposure Mode	: Auto
File Inode Change Date/Time	: 20			Digital Zoom Ratio	: 1
File Permissions	: rw			Focal Length In 35mm Format	: 51 mm
File Type	: JP			Scene Capture Type	: Standard
File Type Extension	: jp			Gain Control	: None
MIME Type	: im			Contrast	: Normal
JFIF Version	: 1			Saturation	: Normal
Exif Byte Order	: Li			Sharpness	: Normal
Make	: NI			Subject Distance Range	: Unknown
Camera Model Name	: NI			GPS Version ID	: 2.3.0.0
Orientation	: Ho			Compression	: JPEG (old-style)
X Resolution	: 30			Thumbnail Offset	: 15340
Y Resolution	: 30			Thumbnail Length	: 6499
Resolution Unit	: in			Rating	: 0
Software	: G1			Date Time	: 2017:08:18 18:59:56
Modify Date	: 20			Date/Time Digitized	: 2017:08:18 18:59:56
Y Cb Cr Positioning	: Ce			Image Width	: 6000
Exposure Time	: 1/			Image Height	: 4000
F Number	: 8			Encoding Process	: Progressive DCT, Huffman coding
Exposure Program	: No			Bits Per Sample	: 8
ISO	: 20			Color Components	: 3
Exif Version	: 02			Y Cb Cr Sub Sampling	: YCbCr4:2:2 (2 1)
Date/Time Original	: 20			Aperture	: 8.0
Create Date	: 20			Auto Focus	: On
Components Configuration	: Y,			Blue Balance	: 1.421875
Compressed Bits Per Pixel	: 2			Image Size	: 6000x4000
Exposure Compensation	: 0			Lens ID	: Unknown (A0 40 2D 53 2C 3C CA 8E)
Max Aperture Value	: 4			Lens	: 18-55mm f/3.5-5.6 G VR [7]
Metering Mode	: M			Megapixels	: 24.0
Light Source	: Un			Red Balance	: 1.996094
Flash	: Av			Scale Factor To 35 mm Equivalent	: 1.5
Focal Length	: 34			Shutter Speed	: 1/250
Maker Note Version	: 2			Create Date	: 2017:08:18 18:59:56.30
Quality	: No			Date/Time Original	: 2017:08:18 18:59:56.30
White Balance	: Au			Modify Date	: 2017:10:02 13:45:16.30
Focus Mode	: AF			Thumbnail Image	: (Binary data 6499 bytes, use -b option to
				Circle Of Confusion	: 0.020 mm
				Depth Of Field	: 5.29 m (2.48 - 7.77 m)
				Field Of View	: 38.6 deg (2.63 m)
				Focal Length	: 34.0 mm (35 mm equivalent: 51.0 mm)
				Hyperfocal Distance	: 7.21 m



- Slika 1, Slika 2 - usporedba

APP1

Dužina EXIF zapisa

DSC_0066-1.jpg	FF	D8	FF	E1	FF	FE	45	78	69	66	00	00	49	49	2A	Exif..II*
00000000	00	08	00	00	00	0B	00	0F	01	02	00	12	00	00	00	
0000000f	94	00	00	00	10	01	02	00	0C	00	00	00	A8	00	00	
0000001e	00	12	01	03	00	01	00	00	00	01	00	00	00	1A	01	
0000002d	05	00	01	00	00	00	B4	00	00	00	1B	01	05	00	01	
0000003c	00	00	00	BC	00	00	00	28	01	03	00	01	00	00	00	 (.....
0000004b	02	00	00	00	31	01	02	00	0A	00	00	00	C4	00	00	1.....
0000005a	00	32	01	02	00	14	00	00	00	00	00	00	13	02	00	
00000069																

Slika 1

APP1

Dužina EXIF zapisa

DSC_0066-2.jpg	FF	D8	FF	E0	00	10	4A	46	49	46	00	01	01	01	01	JFIF.....
00000000	2C	01	2C	00	00	FF	E1	55	39	45	78	69	66	00	00	U9Exif..
0000000f	49	49	2A	00	08	00	00	00	0B	00	0F	01	02	00	12	II*.....
0000001e	00	00	00	92	00	00	00	10	01	02	00	0C	00	00	00	
0000002d	A4	00	00	00	12	01	03	00	01	00	00	00	01	00	00	
0000003c	00	1A	01	05	00	01	00	00	00	B0	00	00	00	1B	01	
0000004b	05	00	01	00	00	00	B8	00	00	00	28	01	03	00	01	 (.....
0000005a	00	00	00	02	00	00	00	31	01	02	00	0C	00	00	00	1.....
00000069																

Slika 2

● Slika 1, Slika 2 - usporedba

DSC_0066-2.jpg

CFA Pa	000056d6	64	66	3A	44	65	73	63	72	69	70	74	69	6F	6E	20	78	6D	6C	6E	df:Description xmln
Custom	000056e9	73	3A	65	78	69	66	3D	27	68	74	74	70	3A	2F	2F	6E	73	2E	61	s:exif='http://ns.a
Exposu	000056fc	64	6F	62	65	2E	63	6F	6D	2F	65	78	69	66	2F	31	2E	30	2F	27	dobe.com/exif/1.0/'
Digita	0000570f	3E	0A	20	20	3C	65	78	69	66	3A	4D	61	6B	65	3E	4E	49	4B	4F	>. <exif:Make>NIKO
Focal	00005722	4E	20	43	4F	52	50	4F	52	41	54	49	4F	4E	3C	2F	65	78	69	66	N CORPORATION</exif
Scene	00005735	3A	4D	61	6B	65	3E	0A	20	20	3C	65	78	69	66	3A	4D	6F	64	65	:Make>. <exif:Mode
Gain C	00005748	6C	3E	4E	49	4B	4F	4E	20	44	35	33	30	30	3C	2F	65	78	69	66	l>NIKON D5300</exif
Contra	0000575b	3A	4D	6F	64	65	6C	3E	0A	20	20	3C	65	78	69	66	3A	4F	72	69	:Model>. <exif:Ori
Satura	0000576e	65	6E	74	61	74	69	6F	6E	3E	54	6F	70	2D	6C	65	66	74	3C	2F	entation>Top-left</
Sharpn	00005781	65	78	69	66	3A	4F	72	69	65	6E	74	61	74	69	6F	6E	3E	0A	20	exif:Orientation>.
Subjec	00005794	20	3C	65	78	69	66	3A	58	52	65	73	6F	6C	75	74	69	6F	6E	3E	<exif:XResolution>
GPS Ve	000057a7	33	30	30	3C	2F	65	78	69	66	3A	58	52	65	73	6F	6C	75	74	69	300</exif:XResoluti
Compre	000057ba	6F	6E	3E	0A	20	20	3C	65	78	69	66	3A	59	52	65	73	6F	6C	75	on>. <exif:YResolu
Thumbn	000057cd	74	69	6F	6E	3E	33	30	30	3C	2F	65	78	69	66	3A	59	52	65	73	tion>300</exif:YRes
Thumbn	000057e0	6F	6C	75	74	69	6F	6E	3E	0A	20	20	3C	65	78	69	66	3A	52	65	olution>. <exif:Re
Rating	000057f3	73	6F	6C	75	74	69	6F	6E	55	6E	69	74	3E	49	6E	63	68	3C	2F	solutionUnit>Inch</
MPF Ve	00005806	65	78	69	66	3A	52	65	73	6F	6C	75	74	69	6F	6E	55	6E	69	74	exif:ResolutionUnit
Number	00005819	3E	0A	20	20	3C	65	78	69	66	3A	53	6F	66	74	77	61	72	65	3E	>. <exif:Software>
MP Ima	0000582c	56	65	72	2E	31	2E	30	31	20	3C	2F	65	78	69	66	3A	53	6F	66	Ver.1.01 </exif:Sof
MP Ima	0000583f	74	77	61	72	65	3E	0A	20	20	3C	65	78	69	66	3A	44	61	74	65	ware>. <exif:Date
MP Ima	00005852	54	69	6D	65	3E	32	30	31	37	3A	30	38	3A	31	38	20	31	38	3A	Time>2017:08:18:1
MP Ima	00005865	35	39	3A	35	36	3C	2F	65	78	69	66	3A	44	61	74	65	54	69	6D	59:56</exif:DateTim
Depend	00005878	65	3E	0A	20	20	3C	65	78	69	66	3A	59	43	62	43	72	50	6F	73	e>. <exif:YCbCrPos

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EXIF zapis slike 2: plaintext (u slici 1 je kodiran)